

Day 03 Python Cheat Sheet

- **Function:** A reusable block of code that performs a specific task.
- **def:** Keyword used to define a function.
- **Parameters:** Variables defined in the function declaration.
- **Arguments:** Actual values passed to a function during a function call.
- **return:** Sends a value back to the caller.
- **Default Arguments:** Uses a default value if no argument is provided.
- **Keyword Arguments:** Pass values using parameter names.
- ***args:** Accepts multiple positional arguments as a tuple.
- ****kwargs:** Accepts multiple keyword arguments as a dictionary.
- **Lambda Function:** A small anonymous function written using the lambda keyword.
- **Recursion:** A function that calls itself until a base condition is met.
- **Module:** A Python file containing reusable code.
- **import:** Imports an entire module.
- **from ... import:** Imports specific functions or objects from a module.
- **math Module:** Used for mathematical operations such as `sqrt()`, `pow()`, `pi`, `ceil()`, and `floor()`.
- **random Module:** Generates random numbers and selects random items.
- **datetime Module:** Works with the current date and time.
- **try:** Contains code that may raise an exception.
- **except:** Handles exceptions if they occur.
- **else:** Executes only if no exception occurs.
- **finally:** Always executes, whether an exception occurs or not.
- **raise:** Manually generates an exception.
- **ValueError:** Raised when an invalid value is provided.
- **ZeroDivisionError:** Raised when dividing by zero.
- **TypeError:** Raised when an operation is performed on incompatible data types.
- **Function Benefits:** Reusability, readability, modularity, and easier maintenance.